Christophe de Dinechin

Software as a Passion

SYSTEMS · LANGUAGES · TOOLS · GAMES · RESEARCH · COMMUNICATION SKILLS



ABOUT

Pushing the boundaries by writing highly innovative and differentiating software that makes a real difference.

French software developer — operating systems, languages, games

Amateur physics researcher and tinkerer in strange forms of art

KEY SKILLS

- ✓ Systems & Virtualization
- ✓ Languages & Compilers
- ✓ Tools & Apps
- ✓ Games & 3D
- ✓ Research
- ✓ Communication Skills

INFO

★ MS Computer Science, École des Mines (1991, Paris)

✉ christophe@dinechin.org

⌘ github.com/c3d

◉ c3d.github.io

TOOLS & TECHNOLOGIES

C · C++ · Virtualization · Kernel · Real-time · 3D · AI · SoC · Validation · Performance · Linux · macOS · OpenShift · RPL · XL · Tao3D · ELFE · SPICE · Make

Interactive constellation

https://c3d.github.io/cv/

SYSTEMS & VIRTUALIZATION

HPVM - Enterprise-grade Itanium virtualization — 2000-2010

- From first line of code and pre-silicon bring up all the way to 7 digits of revenue per quarter

HP ECUTEST Real-Time System — 1994-1998

- Designed HP ECUTEST for engine control unit testing and system validation

SPICE Remote Streaming — 2017-Present

- Part of the Spice team at Red Hat

DxO ONE Connected Camera — 2015-2016

- Dual-core ARM + ISP real-time system

LANGUAGES & COMPILERS

HP aC++ Compiler — 1998-2000

- Wrote the first native Itanium C++ compiler, pre-silicon testing on ski, overriding HP's plans to ship a PA-RISC to Itanium cross-compiler running in the Aries PA-RISC to Itanium dynamic translator
- Exception handling (reference implementation, WIESS paper), C++ ABI initiator, C++ standards committee
- Performance (symbol table, precompiled headers) and build system harmonised across 16 teams

DB48X - Modern RPL Calculator — 2022-Present

- Best scientific calculator today, a modern rebuild of high-end HP-48 and HP-50 calculators

Tao3D — 2010-Present

- Programming language for interactive 3D, derived from XLR

XL — 1992-Present

- Renamed from LX after discussions notably with Daveed Vandevoorde; focus shifted from experimental to extensible

TOOLS & APPS

Make-It-Quick

- Short readable makefiles without automake/cmake complexity

Recorder

- Continuous recording with printf-like statements for performance and debug traces

Emacs on Aqua

- Adapted an old NeXT port to Aqua when only Terminal Emacs shipped

GAMES & 3D

Alpha Waves / Continuum — 1989-1990

- Started in 1988, published by Infogrames in 1990

HP-48 Games — 1989-1993

- PacMan — first use of HP48 hardware scroll (per Paul Courbis)

Taodyne — 2010-2016

- Created Tao3D to present in real-time 3D (big data, movies, glasses-free 3D)

RESEARCH

Computer science research

- Published CS research: WIESS 2000 C++ exception handling; FOSDEM 2024 / FOSDEM 2023 talks on DB48X

12 patents awarded, 3 pending

- 12 granted US patents (HP and Red Hat, virtualization and confidential computing), 3 pending applications including split host/tenant cluster credentials — Google Patents (de Dinechin / Dupont de Dinechin)

Physics research

- A Theory of Incomplete Measurements — unifying GR and QM (TIM, Réunifions la physique)